MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT CONTROLLER OF MINES UNDER GEOLOGY & MINING, COMMERCE & INDUSTRIES DEPARTMENT, GOVERNMENT OF MIZORAM.

(MARCH-2026)

TECHNICAL PAPER-III

(Use of Scienctific calculator is permitted)

Time Allowed : 2 hours FM : 200

PART - A (Multiple Choice Questions) (126 Marks)

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

1. If the "Coefficient of Variation" of ore grade in two different blocks is 15% and 30% respectively, which block has higher relative variability?

(a) The block with 15%

(b) The block with 30%

(c) Both have equal variability

(d) Coefficient of variation does not measure variability

2. The "Poisson Distribution" is most commonly used in mining safety analysis to model:

(a) The average grade of a massive deposit.

(b) The thickness of a coal seam.

(c) The occurrence of rare events, such as underground accidents.

(d) The distribution of particle sizes in mineral dressing.

3. Which measure of central tendency is most affected by extreme values (outliers) in a sampling dataset?

(a) Median (b) Mode

(c) Arithmetic Mean (d) Range

4. An "Exponential Distribution" is frequently used in mine reliability engineering to represent:

(a) Time between equipment failures. (b) The shape of a blast fragment.

(c) The depth of a borehole. (d) The market price of minerals.

5. Heavy Media Separation (HMS) is primarily based on the difference in _____ of the minerals.

(a) Particle Size (b) Magnetic Susceptibility

(c) Specific Gravity (Density) (d) Surface Wettability

6. Jigging is a process of separating particles based on their density using:

(a) A constant horizontal flow of air.
(b) A vertical pulsating column of water.
(c) High-tension electrical wires.
(d) Chemical reagents.

7. Which reagent is commonly used as a 'Frother' in the flotation of sulphide ores?

(a) Xanthates
(b) Pine Oil
(c) Sodium Cyanide
(d) Copper Sulphate

8. According to the Mines Rules, 1955, a First-Aid room must be provided and maintained if the number of persons employed in a mine is more than:

(a) 50
(b) 100
(c) 150
(d) 250

9. As per the Mines Act, 1952, no adult worker shall be required or allowed to work in a mine above ground for more than how many hours in any week?

(a) 40 hours
(b) 45 hours
(c) 48 hours
(d) 54 hours

10. The District Mineral Foundation (DMF) was introduced through an amendment to which of the following Acts?

(a) The Mines Act, 1952
(b) The MMDR Act, 1957
(c) The Mineral Concession Rules, 1960
(d) National Mineral Policy, 2019

11. Which of the following authorities is responsible for granting licenses for the bulk storage of explosives under the Explosives Rules, 2008?

(a) DGMS
(b) Indian Bureau of Mines (IBM)
(c) Petroleum and Explosives Safety Organization (PESO)
(d) State Pollution Control Board

12. According to the Metalliferous Mines Regulations, a "Reportable Injury" means any injury which involves the enforced absence of the injured person from work for a period of:

(a) 24 hours or more
(b) 48 hours or more
(c) 72 hours or more
(d) 7 days or more

13. As per the Mines Rules, 1955, what is the minimum scale of drinking water to be provided per person employed in a mine?

(a) 1 litre
(b) 2 litres
(c) 3 litres
(d) 5 litres

14. According to the National Mineral Policy 2019, which of the following is highly emphasized?

(a) Banning all underground mining
(b) Sustainable mining and Zero Waste mining
(c) Exporting all raw minerals
(d) Nationalizing all private mines

15. Under the provisions of the *Mines and Minerals (Development and Regulation) Amendment Act, 2015*, what is the standard period for which a mining lease for major minerals is now granted?

(a) 20 years
(b) 30 years
(c) 50 years
(d) 99 years

16. Which explosive is most commonly used in open-cast metalliferous mines due to its safety and cost-effectiveness?

(a) Dynamite
(b) Gunpowder
(c) ANFO (Ammonium Nitrate Fuel Oil)
(d) TNT

17. A "Serious Bodily Injury" under the Mines Act involves the permanent loss of any part of the body, or loss of sight/hearing, or fracture of any:

(a) Bone or joint
(b) Rib only
(c) Finger only
(d) Skin tissue

18. Who is held ultimately responsible for the overall financial and safety compliance of the mine under the Mines Act?

(a) The Mine Foreman
(b) The Manager
(c) The Owner / Agent
(d) The Blaster

19. Under the Oilfields (Regulation and Development) Act, 1948, "mineral oils" include:

(a) Coal bed methane only
(b) Natural gas and petroleum
(c) Edible oils extracted near mines
(d) Synthetic lubricants

20. Karl Pearson's coefficient of skewness is zero when:

(a) Mean $\neq$ Mode
(b) Mean = Mode
(c) Mean > Mode
(d) Mode > Mean

21. Coefficient of variation is a measure of:

(a) Central tendency
(b) Absolute dispersion
(c) Relative dispersion
(d) Skewness

22. What is the present value of Rs. 500 receivable at the end of 5 years, if the discount rate is 10% per annum?

(a) 210
(b) 310
(c) 410
(d) 510

23. In a Poisson distribution, the mean and variance are:

(a) Equal
(b) Mean > Variance
(c) Mean < Variance
(d) Unrelated

24. The coefficient of variation is given by:

(a) $\frac{M}{\sigma} \times 100$
(b) $\frac{\sigma}{M} \times 100$
(c) $\frac{\sigma}{N} \times 100$
(d) $M.N \times 100$

25. An equipment costs Rs. 5,00,000. Salvage value after 10 years is Rs. 50,000.

Yearly depreciation under straight-line method is:

(a) ₹ 45,000 (b) ₹ 50,000

(c) ₹ 55,000 (d) ₹ 60,000

26. Primary crushers are mainly classified into:

(a) Jaw and Cone crusher (b) Jaw and Gyratory crusher

(c) Roll and Hammer crusher (d) Cone and Roll crusher

27. According to Stokes' law, settling velocity in hindered settling is:

(a) $\frac{g.\ d^2\left(\rho_s-\rho_p\right)}{18\eta}$ (b) $\frac{18\eta}{\left(\rho_s-\rho_p\right)}$

(c) $\sqrt{3\,g.d^2}$ (d) $\frac{g}{d^2}$

28. The percentage of the mineral occurring as a free particle in the ore in relation to the total content is known as:

(a) Degree of freedom (b) Degree of bondage

(c) Degree of liberation (d) Degree of integration

29. Selection of crusher is done on the basis of particle:

(a) Shape (b) Specific gravity

(c) Size (d) Percentage of mineral in ore

30. Washing is normally performed after:

(a) Grinding (b) Flotation

(c) Primary crushing (d) Smelting

31. What is the minimum No. of persons to be employed when the Owner, Agent or Manager shall constitute a Safety Committee for promoting safety in the mines?

(a) 100 (b) 200

(c) 300 (d) 500

32. Which section of Mines Act 1952 prohibits the employment of women in underground mines?

(a) 45 (b) 46

(c) 47 (d) 48

33. Under Section 5 of the Explosive Substances Act, 1908, possession of explosives under suspicious circumstances may lead to imprisonment up to:

(a) 5 years (b) 10 years

(c) 14 years (d) Life imprisonment.

34. Under National Mineral Policy, 2019, the State acts as:

(a) Absolute owner of mineral resource

(b) Regulatory authority without fiduciary role

(c) Trustee of mineral resources on behalf of the people

(d) Revenue collecting agency only

35. District Mineral Foundation (DMF) is guided by which scheme for benefit devolution?

(a) National Mineral Exploration Trust (NMET) (b) PMKKKY

(c) MCDR 2017 (d) Fifth Schedule

36. Under the Petroleum and Natural Gas Rules, 1959, no person shall prospect for petroleum except under:

(a) Mining Lease (b) Reconnaissance Permit

(c) Petroleum Exploration License (d) Prospecting Certificate

37. As Per the Mizoram Minor Concession Rule, 2000 the maximum ordinary period for which a mining lease may be granted under these rules is:

(a) 2 years (b) 3 years

(c) 4 years (d) 5 years

38. As Per the Mizoram Minor Concession Rule, 2000 the lessee shall pay royalty on minor minerals at the rate specified in the:

(a) First Schedule (b) Second Schedule

(c) Third Schedule (d) Gazette Notification

APTITUDE TEST

39. For doing a certain job, a boy is offered Rs. 5 on the first day. Rs. 15 on the second day, Rs. 45 on the third day and so on. How much will the boy earn at the end of seven days?

(a) Rs. 5486 (b) Rs. 5484

(c) Rs. 5465 (d) Rs. 5450

40. In the sequence of numbers 5, 8, 13, x, 34, 55, 89. The value of x is:

(a) 20 (b) 21

(c) 23 (d) 29

41. If the radius of sphere is decreased by 24%, by what per cent does the surface area decrease?

(a) 44% (b) 49%

(c) 42.24% (d) 46.2%

42. Which one of the following has a greater perimeter than the rest?

(a) A square with an area of 36 sq cm

(b) An equilateral triangle with a side of 9 cm

(c) A rectangle with 10 cm as length and 40 sq cm as area

(d) A circle with a radius of 4 cm

43. 100 students appeared for two examinations, 60 passed the first, 50 passed the second and 30 passed both. The probability that a student selected at random has failed in both the examination, is:

(a) 0.3 (b) 0.2

(c) 0.4 (d) 0.1

44. What sum lent out at simple interest will amount to Rs. 3400 in 3 years at 1% per month?

(a) Rs. 2400 (b) Rs. 1800

(c) Rs. 1600 (d) Rs. 2500

45. Find the odd number out in the following set: 8, 27, 64, 100, 125, 216.

(a) 8 (b) 27

(c) 125 (d) 100

46. Find the wrong number in the series: 3, 5, 9, 15, 25, 33, 45.

(a) 33 (b) 15

(c) 9 (d) 25

47. Choose the word most nearly opposite in meaning to ABSTEMIOUS:

(a) Self-indulgent (b) Frugal

(c) Temperate (d) Moderate

48. Antonym of ENERVATE:

(a) Weaken (b) Exhaust

(c) Invigorate (d) Comfort

49. Analogy — LURID: DISMAL::

(a) Lucid: Obscure (b) Gaudy: Drab

(c) Adroit: Skilful (d) Ephemeral: Eternal

50. Synonym of 'OBSTINATE':

(a) Flexible (b) Humble

(c) Cheerful (d) Stubborn

51. One word for 'fear of heights':

(a) Agrophobia (b) Acrophobia

(c) Hydrophobia (d) Necrophobia

52. "To burn the candle at both ends" means:

(a) To be very resourceful.

(b) To waste energy on a useless project.

(c) To work or play excessively without enough rest.

(d) To light a room efficiently.

53. What is the meaning of the idiom "To tilt at windmills"?

(a) To attack imaginary enemies or grievances.

(b) To engage in agriculture.

(c) To build renewable energy sources.

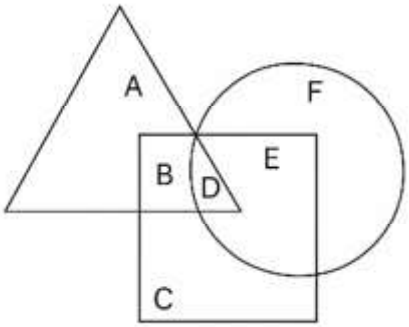
(d) To behave in a strange manner.

54. If you are facing North-West and turn 225 degrees clockwise, which direction are you facing?

(a) East (b) South-West

(c) North (d) South

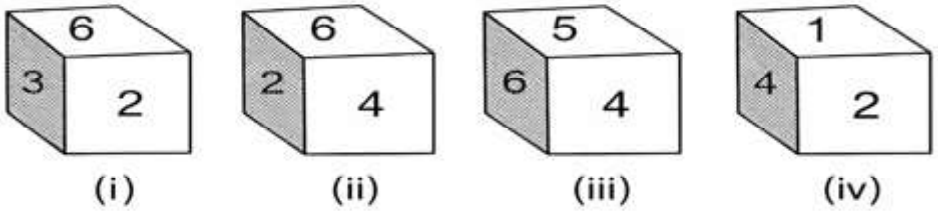
55. In the given figure, the triangle represents girls, the square represents sport persons and the circle represents coaches. The portion in the figure which represents girls who are sports persons but not coaches is the one labelled as



(a) A (b) B

(c) D (d) E

56. Consider the following pictures of dice:

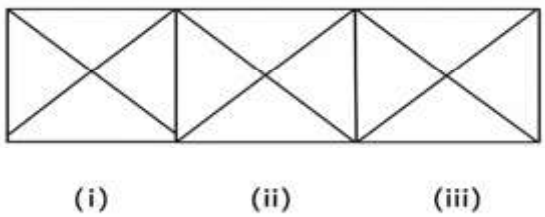


What is the number opposite to 3?

(a) 1 (b) 4

(c) 5 (d) None

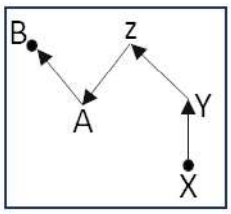
57. How many different triangles are there in the figure shown below?

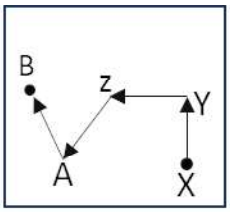


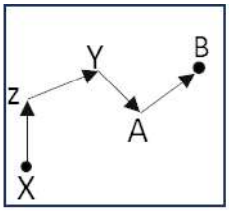
(a) 28 (b) 24

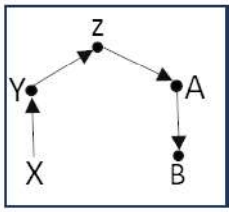
(c) 20 (d) 16

58. Siama starts from a place X towards North and reaches place Y. From there he turns towards North-West and reaches place Z. He then turns towards South-West and walks to place A. From there he turns towards North-West and finally reaches point B. Which of the following figures shows the movement of Siama?

(a) 

(b) 

(c) 

(d) 

59. If ROAD=URDG, then TOWN=?

(a) WSAS (b) WSRA

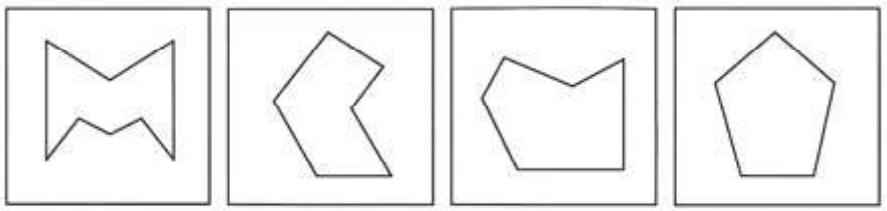
(c) ZRBS (d) WQRA

60. Find the odd one out

(a) AK (b) CM

(c) EP (d) DN

61. Find the odd figure out

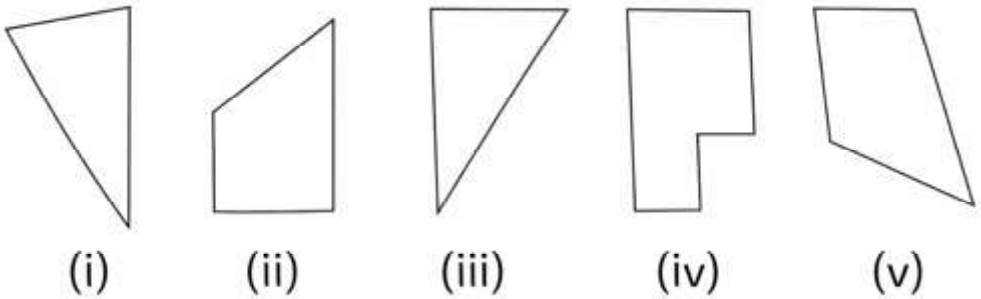


1 **2** **3** **4**

(a) 1 (b) 2

(c) 3 (d) 4

62. Which of the following figures can be joined to form a rectangle?

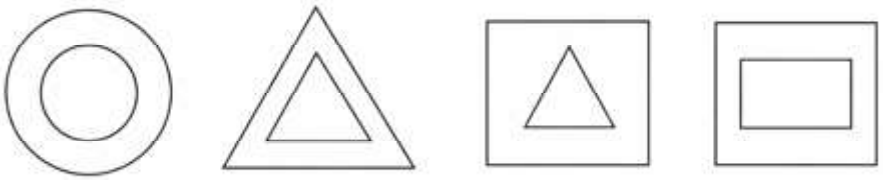


(i) (ii) (iii) (iv) (v)

(a) i, ii, iii (b) ii, iii, iv

(c) iii, iv, v (d) i, iii, v

63. Which of the following figure is different from others?



1 **2** **3** **4**

(a) 1 (b) 2

(c) 3 (d) 4

PART- B *(Conventional Type) (74 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it. Attempt all questions.

1. Explain the difference between Net Present Value (NPV) and Internal Rate of Return (IRR). **(5)**

2. In a geochemical survey, the following values of copper (in ppm) were recorded: 45, 52, 48, 60 and 55. Calculate the Arithmetic Mean and the Standard Deviation for this dataset. **(5)**

3. A mining company plans to set aside ₹ 20,00,000 today for the future reclamation of a mine site. If the annual interest rate is 10%, calculate the Future Worth of this amount at the end of 4 years, compounded annually. **(5)**

4. Briefly explain the working principle of a Ball Mill. Why is it preferred for fine grinding compared to a Rod Mill? **(5)**

5. A primary Jaw Crusher receives a feed where the largest boulder is 800 mm in size. The crusher is set to produce a product with a maximum size of 100 mm. **(5)**

(i) Calculate the Reduction Ratio of this crusher.

(ii) If the reduction ratio must be maintained at 5:1, what would be the required product size for the same feed?

6. Differentiate between "Major Minerals" and "Minor Minerals" under the MMDR Act, 1957. Give two examples of each. **(5)**

7. Briefly outline the basic safety precautions that must be strictly followed during the storage and transport of explosives in a mine, referencing the Explosives Rules, 2008. **(5)**

8. Explain the role and primary functions of the Directorate General of Mines Safety (DGMS) in regulating safety in Indian mines. **(5)**

9. In the context of minor mineral concessions in Mizoram, define "Royalty" and "Dead Rent". What is the standard rule regarding the payment of these two charges by a leaseholder? **(5)**

10. Under the *Metalliferous Mines Regulations (MMR), 1961*, write short notes on: **(2+2=4)**

(i) Competent Person

(ii) Reportable Injury

11. Write a short note on Depreciation, Redemption and Depletion. **(5)**

12. Explain the process of crushing in mineral processing. Differentiate between open-circuit and closed-circuit crushing with the help of a neat flow diagram. **(5)**

13. Discuss the role of Indian Bureau of Mines under MCDR, 2017. **(5)**

14. Explain the general safety provisions under Coal Mines Regulations, 2017. **(5)**

15. Explain the key objectives of National Mineral Policy, 2019. **(5)**

* * * * * * *